Actual public signed Hankel reconstruction

2026-11-13. Source`17fd598fd`. Unsigned evidence [pcc:unverified].

For $|S| = 2a$, $a \geq 2$, actual distinct abscissas $x \in S$, public branches $z(x)^2 = f(x)$ and $\deg f = 3$, define

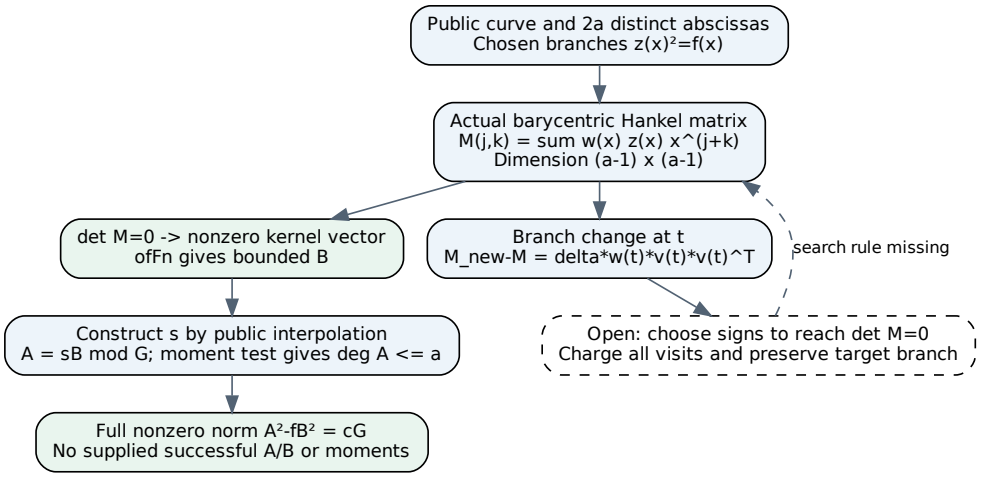
$$G = \prod_{x \in S} (X - x), \quad w_x = \prod_{t \in S \setminus \{x\}} (x - t)^{-1}, \qquad M_{jk} = \sum_{x \in S} w_x z(x) x^{j+k}, \quad 0 \leq j, k < a - 1.$$

This is an actual public symmetric Hankel matrix. Its dimension is $a-1$; its entries depend on the branches, without planted norm polynomials [pcc:unverified].

Exact certificate. Construct s by Lagrange interpolation. For $\deg B < a - 1$, set $A = sB \bmod G$. The finite top-moment test gives

$$\deg A \leq a \quad \Longleftrightarrow \quad M \operatorname{coeff}(B) = 0.$$

If $\det M = 0$, extract a nonzero kernel vector, and use Polynomial.ofFn to construct B and its degree bound. At every actual node, $A = zB$ and $z^2 = f$; therefore $G \mid A^2 - fB^2$. The pole bound caps the norm degree at $2a$. It cannot vanish: $2 \deg A = 3 + 2 \deg B$ is impossible for $B \neq 0$. Thus $A^2 - fB^2 = cG$ with $c \neq 0$. The publicInputCertificate theorem supplies the interpolant itself; no successful A/B, inverse, moment or rank oracle is assumed [pcc:unverified].



Finite moment proof. For $\deg R < \deg G$, the moments $[X^{\deg G - 1}](RX^j \bmod G)$, $j < \deg G - a - 1$, vanish exactly when $\deg R \leq a$. A forbidden highest coefficient is detected by the complementary monomial $X^{\deg G - 1 - \deg R}$. For actual node products, Lagrange's coefficient formula makes these moments the explicit barycentric sums above [pcc:unverified].

Branch update. Changing $z(t)$ by Δ gives $M_{\text{new}} - M = \Delta w_t v_t v_t^T$, $v_t = (1, t, \ldots, t^{a-2})$. A sign flip has $\Delta = -2z(t)$. Arbitrary continuous corrections must still satisfy the curve point equations. The earlier W-jet family has not been mapped to this construction [pcc:unverified].

Fixed controls, evidence and remaining search

Actual finite matrices. Over F17, $f = X^3+2X+2$, use $(0,6),(3,1),(5,1),(6,3),(10,11),(13,7)$. Weights are $[4,9,15,14,16,10]$ [pcc:unverified].

Supplied branches	Hankel matrix	determinant	kernel dim.
Listed points	$\begin{pmatrix} 13 & 15 \\ 15 & 16 \end{pmatrix}$	0	1
Only $(13,7) \to (13,10)$	$\begin{pmatrix} 9 & 14 \\ 14 & 3 \end{pmatrix}$	1	0

The full 6×6 and compressed 2×2 kernels are checked in both directions. Positive reconstruction gives $A = [14,2,7,3]$, $B = [8,1]$, $c = 9$, ascending coefficients, checked against original rows, points, degree bounds and the full norm. No supplied successful A/B enters the solve [pcc:unverified].

All six branch updates for each assignment are checked entrywise. Independent Leibniz determinants also check the cofactor update formula on these finite 2×2 cases. Positive single-flip determinants are $[14,8,16,15,14,1]$. A wrong barycentric weight changes the matrix and rejects the one returned positive candidate; the empty flipped kernel records zero candidate checks. These are fixed identity controls, not independent trials, a sign search or a measured speed comparison [pcc:unverified].

Proof/evidence. Fourteen new statements in NormSignedHankel; whole-study tracked-source gate245 statements,0 additional trusted axioms. Builds001/002 fail;003 passes with a warning;004 removes it;005 supplies the actual public interpolant. A separate fresh replay includes imports; its exact receipt is in checks.json/fresh-status.json. Native tests002:9 passed; strict clippy001 fails on an allocation,002 passes after repair. Preserve all original logs and exits [pcc:unverified].

Public-input claim: `PCC-20261113-e1c10e`.

Degree equivalence: `PCC-20261113-864c3f`.

Literal matrix update: `PCC-20261113-f1012f`.

Original native artifact: `PCC-20261113-a6a482`.

All remain unsigned; default-PATH source-only checks are not_covered, separate from explicit-toolchain kernel checks. Lean4.35.0, CryptoLean `148f5dc389b`, Mathlib`cfc55fc58714`; full source hashes,14-statement proof map, claims and artifact digest are in the source-linked README.

Next obligation and complete cost. Choose the public branch signs so that this actual matrix is singular, keeping a target branch fixed. A direct scan still has 2^{D-1} assignments; polynomial-cost Hankel algebra changes no exponent by itself. The required total remains $\exp(o(\log r))$, including setup, all rejected candidates, divisors and scalar verification. Next test: constrained branch repair and corank/uniqueness structure, rather than assuming a rank gradient. No such search advantage is established here [pcc:unverified].

Primary sources. Zeh–Gentner–Augot1110.3898, equation3 and key equations: GRS weights/Hankel structure are prior art. Cheng cs/0507026, sections3/4: worst-case reductions, not an ordinary-curve average-case bound. Li–Wan–Zhang1501.01138: dense-set and decision assumptions must be charged. Iwata ISAAC2017: independent pair unions differ from dependent transversals. Mahalanobis2607.09814v2: fixed-defect success cost is retained. Exact URLs and read scopes in README. KB tools absent; wrapper exits127. No exhaustive, priority, global impossibility or new ECDLP algorithm claim.